MOHAMED ANSHAF

Python Developer | AI & Full-Stack Development

Mannar , Sri Lanka | +94 71 315 7633 | amanshaf66@gmail.com

LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Python Developer and BSc Information Technology undergraduate (**Open University of Sri Lanka**) with hands-on experience building full-stack web applications and AI-powered solutions using Python, Flask, React.js, PostgreSQL, LangChain and Generative AI. Delivered four end-to-end projects spanning full-stack web development, relational database design, machine learning and Retrieval-Augmented Generation (RAG). Applies Git-based version control and clean, modular design to produce maintainable, production-minded code.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java (Fundamentals)

Backend & Web: Flask, React.js, HTML5, CSS3, Responsive Design, API development, Agile , Scrum

AI / Machine Learning: Generative AI, Large Language Models (LLMs), RAG, LangChain, Gemini API, Scikit-learn, TensorFlow, Google Colab

ML Ops : Azure, Flask, Fast API , Git , GitHub , Linux , Docker (Fundamental)

Databases: PostgreSQL, SQL, Relational Database Design, CRUD Operations

Tools & Practices: VS Code , MySQL, Version Control, Data Analysis, Power BI, Microsoft Excel

PROJECTS

AI-Powered Local Tourist Guide Chatbot |*Python, LangChain, Gemini API, Generative AI*

- Developed an intelligent conversational tour guide assistant designed to eliminate information fragmentation for travelers by providing real-time, accurate local travel guidance.

How It Solves Real-World Problems:

- *Centralizes Fragmented Information*: Replaces scattered websites and blogs by consolidating heritage details, entry fees, visiting hours, and geographical coordinates into a single interactive interface.
- *Prevents AI Hallucinations*: Directs the LLM using **In-Context Knowledge Base Injection** (embedding places.txt directly into system_instruction), ensuring factual accuracy for local landmarks without external vector database dependencies.
- *24/7 Digital Concierge*: Offers an instant, cost-effective, and context-aware travel assistant accessible anytime.

Frameworks, Libraries, and Tools Used

- **AI & LLM Core:** Google GenAI SDK (google-genai), leveraging the lightweight and fast gemini-2.5-flash model for real-time natural language understanding and text generation.
- **Frontend & UI Framework: Streamlit**, enabling a reactive, user-friendly web interface with persistent chat message history and session state management.
- **Backend Logic & State Management:** Python, utilizing script structuring, dynamic file loading, and multi-turn conversation states initialized via (client.chats.create).
- **Knowledge Base:** Custom delimiter-separated flat-file database (places.txt) structured as Name|Latitude|Longitude|Description for lightweight in-context knowledge grounding.

RAG Question-Answering Application | *Python, LangChain, Gemini API, Generative AI*

- Architected a Retrieval-Augmented Generation application that answers natural-language questions from custom document data, grounding LLM responses in source content rather than model memory alone.
- Integrated an LLM API with a document-retrieval pipeline, enabling accurate, context-aware answers over private data without model retraining.

Full-Stack CRUD Web Application | *Python, Flask, PostgreSQL, SQL*

- Designed and managed a PostgreSQL database with SQL queries, ensuring reliable data storage and retrieval across the application.

House Price Prediction Model | *Python, Scikit-learn, Google Colab*

- Developed a supervised machine learning regression model with Scikit-learn to predict house prices from property features, translating raw data into actionable price estimates.

Hospital Landing Page | *React.js, JavaScript, HTML5, CSS3*

- Built a responsive hospital landing page in React.js, ensuring a consistent user experience across devices through reusable, component-based design.

EDUCATION

B.Sc. in Information Technology (Undergraduate)
Open University of Sri Lanka

GCE Advanced Level, Mathematics Stream 2021

CERTIFICATIONS

- Data Analysis with Python, IBM
  https://courses.cognitiveclass.ai/certificates/143e93745d874177819170f9ea0f7f23
- SQL and Relational Database
  https://courses.cognitiveclass.ai/certificates/37dadeac7e274ef18bd5cc8cb7662ada
- Data Science Foundation, Simplilearn [Certificate No: 94819814]

- Artificial Intelligence (1 Week challenge) [NoviTech R&D Certificate ID: NT_5AIV851]